



10103 - The Complex Organic Content of Prestellar Cores

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
NIRCam imaging				
	2	NIRCAM imaging - MODULE B IN MSZ	NIRCam Imaging	(3) L1544
NIRSpec ice absorption spectroscopy				
	3	L1544 Star #3	NIRSpec Fixed Slit Spectroscopy	(1) L1544-STAR3
	8	L1544 Star #1	NIRSpec Fixed Slit Spectroscopy	(4) L1544-STAR1
	9	L1544 Star #5	NIRSpec Fixed Slit Spectroscopy	(5) L1544-STAR5
	10	L1544 Star #4	NIRSpec Fixed Slit Spectroscopy	(2) L1544-STAR4
MIRI ice absorption spectroscopy				
	5	L1544 Star #3	MIRI Medium Resolution Spectroscopy	(1) L1544-STAR3
	6	L1544 Star #1	MIRI Medium Resolution Spectroscopy	(4) L1544-STAR1
	7	L1544 Star #5	MIRI Medium Resolution Spectroscopy	(5) L1544-STAR5
	11	L1544 Star #4	MIRI Medium Resolution Spectroscopy	(2) L1544-STAR4

ABSTRACT

Prestellar cores represent the precursors of Solar-type systems and as such, their chemical composition influences the final chemical content of protoplanetary disks and of the comets that form within them. Complex organic molecules (COMs), including potentially the building blocks of proteins and RNA/DNA, may be inherited by comets and meteorites from the parent cloud. Thus, a better understanding of the solid COM inventory of prestellar cores can yield unique insights into the prebiotic chemistry of early Earth. JWST has characterized the COM ice chemical composition (CH₃OH and larger COMs) in molecular clouds before and after the prestellar core stage. But no information is available about the solid COM content in prestellar cores, which set the COM chemical composition inherited by comets. We propose to carry out high-sensitivity spectroscopy with JWST to measure the COM ice content in the best-studied prestellar core, L1544, toward four background stars with previous CH₃OH ground-based observations and that probe extinctions between 9 and 27 mag. Combined with SPHEREx data (extinctions <9 mag), these observations will investigate the transition from the apolar ice phase, where CO starts depleting, to the CH₃OH ice phase, where CH₃OH and larger COMs form. JWST is unique in its ability to study cosmic ice, though it is still underutilized for prestellar cores. Hence, we propose this project on L1544 which, together with JWST archival data, will establish how COMs (especially large COMs) form, grow and evolve along the process of star and planetary formation, from molecular clouds, to prestellar cores, protostars, protoplanetary disks and comets.

OBSERVING DESCRIPTION

The main goal of this proposal is to measure the abundance of complex organic molecules (COMs) in dust-grain ice mantles toward the L1544 prestellar core at different extinctions to constrain COM formation as a function of gas-phase CO depletion and CO ice coverage. For this, we will carry out ice absorption spectroscopy toward four background stars in this core, Stars #1, #3, #4 and #5 (with visual extinctions $A_v = 27, 19, 9$ and 12 mag, respectively), where observations of methanol ice, the parent species of more complex COMs, have already been carried out (Goto et al. 2021). These JWST observations will be complemented by SPHEREx ice observations obtained toward L1544, which will probe visual extinctions lower than $A_v=9$ mag. By probing different extinctions, we can study large COM production in ices as a function of varying H₂O/CO mixtures.

We will carry out fixed-slit spectroscopy with NIRSpec to obtain ice absorption spectra between 1.6 and 5.3 microns to measure the ice content in water (H₂O), carbon monoxide (CO), carbon dioxide (CO₂), and methanol (CH₃OH; the simplest COM). Other COMs such as acetaldehyde (CH₃CHO), and methyl formate (CH₃OCHO), are predicted to be formed on grain surfaces in models of the chemistry of COMs in prestellar cores. Therefore, we will use the Medium-Resolution Spectroscopy (MRS) mode of MIRI to measure the COM ice content toward the background Stars #1, #3, #4 and #5 in L1544 probing different extinctions with different H₂O/CO ice abundance ratios.

The JWST ice spectra will be complemented with laboratory experiments and with models of the chemistry of COMs under cold interstellar conditions, to interpret the JWST results. This will allow us to model the chemistry of this core with unprecedented detail, providing essential information into the properties and history of the ice, the formation of COMs, and the COM reservoir transferred to cometesimals.

Finally, since L1544 shows A_v extinction values up to 100 mag at the core's centre, there are no known background sources toward the densest and innermost regions in the core. Therefore, we also request to perform the imaging of the central region of L1544 with NIRCам to identify additional background stars with $A_v > 50$ mag for future JWST studies.

Proposal 10103 - Targets - The Complex Organic Content of Prestellar Cores

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	L1544-STAR3	RA: 05 04 21.7200 (76.0905000d) Dec: +25 12 8.60 (25.20239d) Equinox: J2000		
Comments: Category=ISM Description=[Interstellar molecules, Molecular clouds] Extended=NO				
(2)	L1544-STAR4	RA: 05 04 12.5000 (76.0520833d) Dec: +25 09 54.20 (25.16506d) Equinox: J2000		
Comments: Category=ISM Description=[Interstellar clouds, Interstellar molecules]				
(3)	L1544	RA: 05 04 16.0000 (76.0666667d) Dec: +25 10 40.00 (25.17778d) Equinox: J2000		
Comments: Category=ISM Description=[Interstellar dust, Molecular clouds]				
(4)	L1544-STAR1	RA: 05 04 15.9100 (76.0662917d) Dec: +25 11 57.30 (25.19925d) Equinox: J2000		
Comments: Category=ISM Description=[Interstellar clouds, Interstellar molecules]				
(5)	L1544-STAR5	RA: 05 04 22.0900 (76.0920417d) Dec: +25 09 37.80 (25.16050d) Equinox: J2000		
Comments: Category=ISM Description=[Interstellar clouds, Interstellar molecules]				

Proposal 10103 - Observation 2 - The Complex Organic Content of Prestellar Cores

Observation	Proposal 10103, Observation 2: NIRCAM imaging - MODULE B IN MSZ									Fri Mar 13 18:05:01 GMT 2026	
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(3)	L1544	RA: 05 04 16.0000 (76.0666667d) Dec: +25 10 40.00 (25.17778d) Equinox: J2000								
	Comments: Category=ISM Description=[Interstellar dust, Molecular clouds]										
Template	Module					Subarray					
	B					FULL					
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEBOX		3		STANDARD				2	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F212N	F360M	RAPID	2	1	6	6	128.841	221350	
	2	F200W	F466N+F444W	RAPID	10	1	6	6	644.206	221350	
Special Requirements	Aperture PA Range 79.24262691 to 93.86262691 Degrees (V3 79.19 to 93.81)										

Proposal 10103 - Observation 3 - The Complex Organic Content of Prestellar Cores

Observation	Proposal 10103, Observation 3: L1544 Star #3											Fri Mar 13 18:05:01 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
	Comments: These observations should be carried out a couple of months after the NIRCам preimaging so that we have enough time to analyse the NIRCам image and select the best background stars to carry out the ice absorption spectroscopy.											
We will use both the S200A1 and S200A2 slits so that the gaps in the filters are fully covered within our observations. Note that the number of groups requested per filter and per source is half the one indicated in the ETC Workbook number 56351, because both S200A1 and S200A2 will be used for each target. The final spectrum containing the S200A1+S200A2 data will reach the requested SNR>160 at 3.54 microns.												
Diagnos	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(1)	L1544-STAR3	RA: 05 04 21.7200 (76.0905000d) Dec: +25 12 8.60 (25.20239d) Equinox: J2000									
	Comments: Category=ISM Description=[Interstellar molecules, Molecular clouds] Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	SAME	WATA	SUB32	F140X	NRSRAPID	3	1	1	0.08	274483	
Template	HFF Readout Mode				Slit				Subarray			
	false				S200A1 and S200A2				ALLSLITS			
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	2						SPATIAL				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Ex p	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395H/F290LP	S200A1	NRS	3	1	1	NONE	4	4	285.77	274483
	2	G395H/F290LP	S200A2	NRS	3	1	2	NONE	4	4	285.77	
	3	G235H/F170LP	S200A2	NRS	6	1	3	NONE	4	4	549.482	
	4	G235H/F170LP	S200A1	NRS	6	1	4	NONE	4	4	549.482	274483

Proposal 10103 - Observation 8 - The Complex Organic Content of Prestellar Cores

Observation	Proposal 10103, Observation 8: L1544 Star #1											Fri Mar 13 18:05:01 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
	Comments: These observations should be carried out a couple of months after the NIRCам preimaging so that we have enough time to analyse the NIRCам image and select the best background stars to carry out the ice absorption spectroscopy.											
We will use both the S200A1 and S200A2 slits so that the gaps in the filters are fully covered within our observations. Note that the number of groups requested per filter and per source is half the one indicated in the ETC Workbook number 56351, because both S200A1 and S200A2 will be used for each target. The final spectrum containing the S200A1+S200A2 data will reach the requested SNR>160 at 3.54 microns.												
Diagnos	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(4)	L1544-STAR1	RA: 05 04 15.9100 (76.0662917d) Dec: +25 11 57.30 (25.19925d) Equinox: J2000									
	Comments: Category=ISM Description=[Interstellar clouds, Interstellar molecules]											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	SAME	WATA	SUB32	F140X	NRSRAPID	3	1	1	0.08	274483	
Template	HFF Readout Mode				Slit				Subarray			
	false				S200A1 and S200A2				ALLSLITS			
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	2						SPATIAL				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Ex p	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395H/F290LP	S200A1	NRS	22	1	1	NONE	4	4	1955.946	274483
	2	G395H/F290LP	S200A2	NRS	22	1	2	NONE	4	4	1955.946	
	3	G235H/F170LP	S200A2	NRS	37	1	3	NONE	4	4	3274.506	
	4	G235H/F170LP	S200A1	NRS	37	1	4	NONE	4	4	3274.506	274483

Proposal 10103 - Observation 9 - The Complex Organic Content of Prestellar Cores

Observation	Proposal 10103, Observation 9: L1544 Star #5											Fri Mar 13 18:05:01 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
	Comments: These observations should be carried out a couple of months after the NIRCам preimaging so that we have enough time to analyse the NIRCам image and select the best background stars to carry out the ice absorption spectroscopy.											
We will use both the S200A1 and S200A2 slits so that the gaps in the filters are fully covered within our observations. Note that the number of groups requested per filter and per source is half the one indicated in the ETC Workbook number 56351, because both S200A1 and S200A2 will be used for each target. The final spectrum containing the S200A1+S200A2 data will reach the requested SNR>160 at 3.54 microns.												
Diagnos	(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(5)	L1544-STAR5	RA: 05 04 22.0900 (76.0920417d) Dec: +25 09 37.80 (25.16050d) Equinox: J2000									
	Comments: Category=ISM Description=[Interstellar clouds, Interstellar molecules]											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	SAME	WATA	SUB32	F140X	NRSRAPID	3	1	1	0.08	274483	
Template	HFF Readout Mode				Slit				Subarray			
	false				S200A1 and S200A2				ALLSLITS			
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	2						SPATIAL				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Ex p	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395H/F290LP	S200A1	NRS	23	1	1	NONE	4	4	2043.85	274483
	2	G395H/F290LP	S200A2	NRS	23	1	2	NONE	4	4	2043.85	
	3	G235H/F170LP	S200A2	NRS	37	1	3	NONE	4	4	3274.506	
	4	G235H/F170LP	S200A1	NRS	37	1	4	NONE	4	4	3274.506	274483

Proposal 10103 - Observation 10 - The Complex Organic Content of Prestellar Cores

Observation	Proposal 10103, Observation 10: L1544 Star #4											Fri Mar 13 18:05:01 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(2)	L1544-STAR4	RA: 05 04 12.5000 (76.0520833d) Dec: +25 09 54.20 (25.16506d) Equinox: J2000									
	Comments: Category=ISM Description=[Interstellar clouds, Interstellar molecules]											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	SAME	WATA	SUB32	F140X	NRSRAPID	3	1	1	0.08	274483	
Template	HFF Readout Mode				Slit				Subarray			
	false				S200A1 and S200A2				ALLSLITS			
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	2						SPATIAL				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395H/F290LP	S200A1	NRS	22	1	1	NONE	4	4	1955.946	274483
	2	G395H/F290LP	S200A2	NRS	22	1	2	NONE	4	4	1955.946	
	3	G235H/F170LP	S200A2	NRS	63	1	3	NONE	4	4	5560.01	
	4	G235H/F170LP	S200A1	NRS	63	1	4	NONE	4	4	5560.01	274483

Proposal 10103 - Observation 5 - The Complex Organic Content of Prestellar Cores

Observation	Proposal 10103, Observation 5: L1544 Star #3												Fri Mar 13 18:05:01 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Comments: These observations should be carried out a couple of months after the NIRC <i>am</i> preimaging so that we have enough time to analyse the NIRC <i>am</i> image and select the best background stars to carry out the ice absorption spectroscopy.												
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(1)	L1544-STAR3	RA: 05 04 21.7200 (76.0905000d) Dec: +25 12 8.60 (25.20239d) Equinox: J2000										
	Comments: Category=ISM Description=[Interstellar molecules, Molecular clouds] Extended=NO												
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID				
	1	SAME	F1000W	FAST	4	1	1	11.1	274483				
Template	Primary Channel		Simultaneous Imaging			Imager Subarray			Grating Wheel Direction				
	All MRS		NO			FULL			Allow Auto Reorder				
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				POSITIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	SHORT(A)	MRSLONG		FASTR1	58	1	1	Dither 1	4	4	643.809	274483
	1	SHORT(A)	MRSSHORT		FASTR1	58	1	1	Dither 1	4	4	643.809	274483
	2	MEDIUM(B)	MRSLONG		FASTR1	37	2	1	Dither 1	4	8	832.512	274483
	2	MEDIUM(B)	MRSSHORT		FASTR1	37	2	1	Dither 1	4	8	832.512	274483
	3	LONG(C)	MRSLONG		FASTR1	50	1	1	Dither 1	4	4	555.008	274483
	3	LONG(C)	MRSSHORT		FASTR1	50	1	1	Dither 1	4	4	555.008	274483

Proposal 10103 - Observation 6 - The Complex Organic Content of Prestellar Cores

Observation	Proposal 10103, Observation 6: L1544 Star #1												Fri Mar 13 18:05:01 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Comments: These observations should be carried out a couple of months after the NIRCcam preimaging so that we have enough time to analyse the NIRCcam image and select the best background stars to carry out the ice absorption spectroscopy.												
We note that the observing visit of the background star with Av=100 mag (visit 6:1) had to be split into two different visits (visits 6:1 and 7:1). This is due to the fact that one single visit gives the following warning message: "Data volume for this visit 41320.82 MB exceeds half the maximum allowed of 66566.23 MB". Therefore, we have opted for spliting the visit 6:1 into two new ones (visits 6:1 and 7:1), but with half the total exposure time for each of them so that this data volume warning is avoided.													
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(4)	L1544-STAR1	RA: 05 04 15.9100 (76.0662917d) Dec: +25 11 57.30 (25.19925d) Equinox: J2000										
	Comments:												
	Category=ISM Description=[Interstellar clouds, Interstellar molecules]												
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID				
	1	SAME	F1000W	FAST	4	1	1	11.1	274483				
Template	Primary Channel		Simultaneous Imaging				Imager Subarray				Grating Wheel Direction		
	All MRS		NO				FULL				Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				POSITIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	SHORT(A)	MRSLONG		SLOWR1	60	1	1	Dither 1	4	4	5733.581	274483
	1	SHORT(A)	MRSSHORT		SLOWR1	60	1	1	Dither 1	4	4	5733.581	274483
	2	MEDIUM(B)	MRSLONG		SLOWR1	35	2	1	Dither 1	4	8	6784.737	274483
	2	MEDIUM(B)	MRSSHORT		SLOWR1	35	2	1	Dither 1	4	8	6784.737	274483
	3	LONG(C)	MRSLONG		SLOWR1	50	1	1	Dither 1	4	4	4777.984	274483
	3	LONG(C)	MRSSHORT		SLOWR1	50	1	1	Dither 1	4	4	4777.984	274483

Proposal 10103 - Observation 7 - The Complex Organic Content of Prestellar Cores

Observation	Proposal 10103, Observation 7: L1544 Star #5												Fri Mar 13 18:05:01 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Comments: These observations should be carried out a couple of months after the NIRCcam preimaging so that we have enough time to analyse the NIRCcam image and select the best background stars to carry out the ice absorption spectroscopy. We note that the observing visit of the background star with Av=100 mag (visit 6:1) had to be split into two different visits (visits 6:1 and 7:1). This is due to the fact that one single visit gives the following warning message: "Data volume for this visit 41320.82 MB exceeds half the maximum allowed of 66566.23 MB". Therefore, we have opted for splitting the visit 6:1 into two new ones (visits 6:1 and 7:1), but with half the total exposure time for each of them so that this data volume warning is avoided.												
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(5)	L1544-STAR5	RA: 05 04 22.0900 (76.0920417d) Dec: +25 09 37.80 (25.16050d) Equinox: J2000										
	Comments: Category=ISM Description=[Interstellar clouds, Interstellar molecules]												
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID				
	1	SAME	F1000W	FAST	4	1	1	11.1	274483				
Template	Primary Channel		Simultaneous Imaging				Imager Subarray				Grating Wheel Direction		
	All MRS		NO				FULL				Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				POSITIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	SHORT(A)	MRSLONG		SLOWR1	50	2	1	Dither 1	4	8	9651.528	274483
	1	SHORT(A)	MRSSHORT		SLOWR1	50	2	1	Dither 1	4	8	9651.528	274483
	2	MEDIUM(B)	MRSLONG		SLOWR1	60	2	1	Dither 1	4	8	11562.721	274483
	2	MEDIUM(B)	MRSSHORT		SLOWR1	60	2	1	Dither 1	4	8	11562.721	274483
	3	LONG(C)	MRSLONG		SLOWR1	45	2	1	Dither 1	4	8	8695.931	274483
	3	LONG(C)	MRSSHORT		SLOWR1	45	2	1	Dither 1	4	8	8695.931	274483

Proposal 10103 - Observation 11 - The Complex Organic Content of Prestellar Cores

Observation	Proposal 10103, Observation 11: L1544 Star #4												Fri Mar 13 18:05:01 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
Diagnostics	(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(2)	L1544-STAR4	RA: 05 04 12.5000 (76.0520833d) Dec: +25 09 54.20 (25.16506d) Equinox: J2000										
	Comments: Category=ISM Description=[Interstellar clouds, Interstellar molecules]												
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID				
	1	SAME	F1000W	FAST	4	1	1	11.1	274483				
Template	Primary Channel		Simultaneous Imaging				Imager Subarray				Grating Wheel Direction		
	All MRS		NO				FULL				Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	SHORT(A)	MRSLONG		SLOWR1	40	1	1	Dither 1	4	4	3822.387	274483
	1	SHORT(A)	MRSSHORT		SLOWR1	40	1	1	Dither 1	4	4	3822.387	274483
	2	MEDIUM(B)	MRSLONG		SLOWR1	25	2	1	Dither 1	4	8	4873.544	274483
	2	MEDIUM(B)	MRSSHORT		SLOWR1	25	2	1	Dither 1	4	8	4873.544	274483
	3	LONG(C)	MRSLONG		SLOWR1	34	1	1	Dither 1	4	4	3249.029	274483
	3	LONG(C)	MRSSHORT		SLOWR1	34	1	1	Dither 1	4	4	3249.029	274483